**1** **(a)** 1680 **[B1]**

**(b)** M1 for $1680 - 1050$ or 630 seen

140 **[A1]**

*[Total Marks: 3]*

**2** $255 \div 0.85$ **[M1]**

\$300 **[A1]**

\$100 **[A1]**

*[Total Marks: 3]*

**3** $(3 \times 1.20) + (4 \times 2.50)$ or $3.60 + 10.00$ or 13.60 **[M1]**

$20 - 13.60$ **[M1]**

6.40 **[A1]**

*[Total Marks: 3]*

**4** $300 \times 1.50 = 450$ **[M1]**

$450 - 210 = 240$ **[M1]**

$240 \div 1.50 = 160$ **[A1]**

*[Total Marks: 3]*

**5** **(a)** Cost of pens $= 4 \times 2.35 = 9.40$ **[M1]**

Cost of notebooks $= 3 \times 4.50 = 13.50$ **[M1]**

Change $= 50 - (9.40 + 13.50) = 27.10$ **[A1]**

**(b)** 54 **[B1]**

*[Total Marks: 4]*

**6** **(a)** $850 \times 148$ **[M1]**

125800 **[A1]**

**(b)** $125800 - (45000 + 31500)$ or 49300 **[M1]**

$\frac{49300}{152}$ or 324.342... **[M1]**

324.34 **[A1]**

*[Total Marks: 5]*

**7** **(a)** $1\,250\,000 \div 145$ **[M1]**

8620.69 **[A1]**

**(b)** Total cost $= 8620.69 + 600 = 9220.69$ **[M1]**

Profit $= 12\,500 - 9220.69 = 3279.31$ (or $\frac{12500 - 9220.69}{9220.69}$ ) **[M1]**

35.6 (accept $35.56\ldots$) **[A1]**

*[Total Marks: 5]*

**8 (a)** $450 \times 1.14$ **[M1]**

513 **[A1]**

**(b)** $1.32 \div 1.14$ **[M1]**

1.1578... **[A1]**

1.16 **[A1]**

*[Total Marks: 5]*

**9 (a)** M1 for $680 \times 1.32$ (implies 897.60)

M1 for their $897.60 - 850$
OR
M1 for $850 \div 1.32$ (implies $643.939\ldots$)
M1 for (680 − their $643.939\ldots$) $\times 1.32$

47.60 (accept 47.6) **[A1]**

**(b)** M1 for $850 \times 1.04$ or $850 \times 0.04 + 850$ (implies 884)

M1 for their $884 \times 0.85$

751.40 (accept 751.4) **[A1]**

*[Total Marks: 6]*

**10 (a)** Amount in euros $= 450 \times 1.18 = 531$ **[M1]**

Remaining euros $= 531 - 285 = 246$ **[M1]**

Amount back in pounds $= 246 \div 1.22$ **[M1]**

201.639... rounding to 201.64 **[A1]**

**(b)** Sale price $= 850 \times 0.85$ $(= 722.5)$ **[M1]**

Final price $= 722.5 \times 0.95 = 686.375$ which rounds to 686.38 **[A1]**

*[Total Marks: 6]*

**11 (a)** $850 \times 0.98$ or $850 - (850 \times 0.02)$ **[M1]**

833 **[A1]**

149940 **[A1]**

**(b)** $149940 - 115000$ $(= 34940)$ **[M1]**

their $34940 - 500$ $(= 34440)$ **[M1]**

their $34440 \div 184$ **[M1]**

187.17 (accept 187.17...) **[A1]**

*[Total Marks: 7]*

**12** **(a)** $\frac{17400 + 1200}{1.24}$ or $\frac{18600}{1.24}$ **[M1]**

15000 **[A1]**

**(b)** $17100 - 15000$ or 2100 **[M1]**

$\frac{2100}{15000} \times 100$ **[M1]**

14 **[A1]**

**(c)** $2100 \times (1 - 0.02)$ or $2100 \times 0.98$ or 2058 **[M1]**

$2058 \times 1.16$ **[M1]**

2387.28 **[A1]**

*[Total Marks: 8]*

**13** Total revenue in USD $= 1000 \times 4.20 = 4200$ **[B1]**

Convert revenue to EUR: $4200 \div 1.20$ $(= 3500)$ **[M1]**

Equate to cost with profit multiplier: Total Cost $\times 1.40 = 3500$ or $3500 \div 1.40$ $(= 2500)$ **[M1]**

Form equation for cost: $1000y + 250 = 2500$ **[M1]**

$y = 2.25$ **[A1]**

*[Total Marks: 5]*

**14** **(a)** $800 \times 0.85$ or 680 **[M1]**

$680 - 420$ or 260 **[M1]**

$260 \div 0.82$ **[M1]**

317.07 **[A1]**

**(b)** $61.20 \div (1 - 0.15)$ or $61.20 \div 0.85$ **[M1]**

72 **[A1]**

*[Total Marks: 6]*

**15** **(a)** Cost of Brand A per 100g $= 12.50 \div 5 = 2.50$ (or per gram $= 0.025$) **[M1]**

Cost of Brand B per 100g $= 8.40 \div 3 = 2.80$ (or per gram $= 0.028$) **[M1]**

Brand A with correct supporting working **[A1]**

**(b)** $x + y = 2000$ **[M1]**

Cost equation: $0.025x + 0.028y = 54.10$ (or equivalent, e.g. $\frac{12.5}{500}x + \frac{8.4}{300}y = 54.1$) **[M1]**

Substitute $y = 2000 - x$: $0.025x + 0.028(2000 - x) = 54.10$ **[M1]**

$0.025x + 56 - 0.028x = 54.10 \implies -0.003x = -1.9 \implies x = 633.3...$ and $y = 1366.6...$

Accept $x = 633$ and $y = 1367$ (or exact fractions $\frac{1900}{3}$ and $\frac{4100}{3}$) **[A1]**

*[Total Marks: 7]*

**16 (a)** $\frac{400}{x}$ **[B1]**

**(b)** $\frac{450}{x - 0.05} - \frac{400}{x} = 100$ **[M1]**

$450x - 400(x - 0.05) = 100x(x - 0.05)$ **[M1]**

$450x - 400x + 20 = 100x^2 - 5x$ **[M1]**

$50x + 20 = 100x^2 - 5x \implies 100x^2 - 55x - 20 = 0$ **[A1]**

Dividing by 5 yields $20x^2 - 11x - 4 = 0$ (must show working to reach final equation) **[A1]**

**(c)** $(5x - 4)(4x + 1) = 0$ or use of quadratic formula $\frac{11 \pm \sqrt{(-11)^2 - 4(20)(-4)}}{2(20)}$ **[M2]**

If M2 not scored, allow [M1] for $(5x \pm a)(4x \pm b) = 0$ where $ab = 4$ or $a(4) \pm b(5) = -11$, or correct substitution into formula but with one error.

0.8 (with $-0.25$ rejected or ignored) **[A1]**

*[Total Marks: 8]*

**17 (a)** M1 for $21500 - 24000x = 740$

M1 for $24000x = 20760$

0.865 **[A1]**

**(b)** M1 for $780 \times 1.28$ (implies 998.4)

M1 for their $998.4 \div (1 - 0.025)$ or their $998.4 \div 0.975$

1024 **[A1]**

**(c)** M1 for setting up a correct equation, e.g. $R \times (1 - 0.08) \times k = R \times 0.828$ or $0.92k = 0.828$

M1 for $k = \frac{0.828}{0.92}$

0.9 **[A1]**

*[Total Marks: 9]*

**18 (a)** $840 \times 1.15$ or 966 **[M1]**

$\frac{821.10}{966}$ or 0.85 **[M1]**

15 **[A1]**

**(b)** $\frac{2430000}{450}$ or 5400 **[M1]**

$\frac{5400}{7.2}$ or 750 **[M1]**

90 **[A1]**

**(c)** $840 \times 1.025$ or 861 **[M1]**

$\frac{861}{1.32}$ or 652.272... **[M1]**

652.27 **[A1]**

*[Total Marks: 9]*